

BANARAS HINDU UNIVERSITY

B.Voc. Semester IV Examination, 2022-23

Paper: BVCA-404 — Microprocessor

Time: Three Hours

Maximum Marks: 70

Instructions: Answer the questions in each section as instructed. All questions carry marks as indicated.

SECTION — A

Long-Answer Questions. Answer any two questions:

 $2 \times 12 = 24$ Marks

Question 1. Explain the architecture of the 8085 microprocessor, describing each component and how they interact with each other.

Question 2. Discuss the various types of addressing modes in the 8085 microprocessor, providing a suitable example for each.

Question 3. Explain the role of the stack and the stack pointer in the 8085 microprocessor. Illustrate your explanation with practical examples.

Question 4. Describe the differences between a microprocessor and a microcontroller, with emphasis on the 8085 as a specific example of a microprocessor.

SECTION — B

Question 5. Attempt any six parts (Answer in about 200 words each): **$6 \times 6 = 36$ Marks**

- (a) Briefly explain the function of the Arithmetic Logic Unit (ALU) in the 8085 microprocessor.
- (b) List and explain the status of any three flags in the 8085 microprocessor's flag register.
- (c) Define the terms "Opcode" and "Operand" with respect to the 8085 microprocessor's instruction format.
- (d) Explain the purpose and classifications of the instruction set in a microprocessor.
- (e) If a program starts at memory location 2000H, and the first instruction is LDA 2050H, what will be the content of the accumulator and the program counter after this instruction is executed?
- (f) Write an assembly language program for the 8085 microprocessor to add two 8-bit numbers stored at memory locations 3000H and 3001H, and store the result in memory location 3002H.
- (g) Given that the stack pointer (SP) is initially at 4000H, what will be its value after executing the following sequence of instructions?

```
PUSH H  
PUSH D  
POP H
```

- (h) Explain the memory address accessed by the instruction sequence:

```
LXI H, 4050H  
MOV A, M
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- (i) What is interrupt-driven I/O and how is it implemented in the 8085 microprocessor?

SECTION — C

Question 6. Attempt all parts:

10 × 1 = 10 Marks

- (a) What does the acronym ALU stand for?
- (b) Name the register that stores the result of an arithmetic or logical operation in the 8085 microprocessor.
- (c) What is the size of the data bus in the 8085 microprocessor?
- (d) Which pin on the 8085 microprocessor is used for a non-maskable hardware interrupt?
- (e) How many machine cycles does the register-to-register MOV instruction require?
- (f) How do you initialize the stack pointer in the 8085 microprocessor?
- (g) What does the HLT instruction do?
- (h) Which flag is set if an arithmetic or logical operation results in zero in the 8085 microprocessor?
- (i) What is the opcode (in hex) for the NOP instruction in the 8085 microprocessor?
- (j) How many bytes does the LDA instruction occupy?